CS1910 Computer Science I • Variables, Literals and Types **Partner workbook**

Drawing Regular Polygons

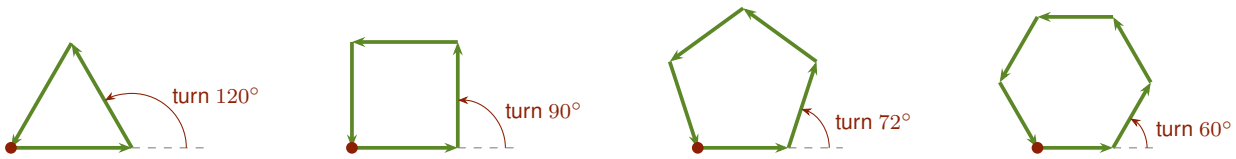
In lecture with a partner • about 10 minutes

Name: _____ **Partner:** _____

This PDF can be filled in: click in an answer space to type, or print it and write. Draw diagrams with your PDF viewer's pen or markup tool, or on paper.

What this workbook practises

In lecture we drew a square with turtle graphics, and then improved the program by storing the side length in a variable instead of repeating a literal. Now you will **reuse** that idea and **generalize** it: first to a triangle, then to *any* regular polygon, a closed shape whose sides are all the same length and whose angles are all equal (pentagon, hexagon, dodecagon, square, . . .).



The turtle starts at the red dot, draws a side, turns, and repeats. The red arc is the turn it makes at each corner.

1 A triangle

about 4 minutes

1. What do you already have from the square program that you can reuse?

2. What do you need to know to draw an equilateral triangle? How many sides does it have, and what is its interior angle?

3. How far does the turtle turn at each corner? Why is it not the interior angle?

4. Write pseudocode to draw a triangle.

2 Any regular polygon

about 4 minutes

5. Fill in the table. What is the pattern?

Shape	triangle	square	pentagon	hexagon	dodecagon
sides	3	4	5	6	12
turn (degrees)	120				

6. Write pseudocode that draws any regular polygon, using a new variable `sides`. If `sides` is 6, it should draw a hexagon.

7. If you finish early: convert your pseudocode to Python, reusing the square program from lecture.

Before you leave

about 1 minutes

- [ ] I can reuse an earlier solution (the square) to solve a new problem.
- [ ] I can explain why the turtle turns $360 \div$ sides degrees.
- [ ] I can use a variable to make one program work for many shapes.